The totally symmetric $\zeta(7)$ cellular family: exact recurrence, two period ladders, the value of P_3, and a Dwork-congruence reduction of the central-binomial law

18 July 2026

Abstract

We report the results of an intensive computational–theoretical campaign on Brown–Zudilin cellular integrals at weights five and seven. For the totally symmetric $\zeta(7)$ family on $\overline{\mathcal{M}}_{0,10}$ we determine the exact minimal recurrence of the leading coefficients q_n (order 4, degree 19, from 105 computed terms; certified against all 74 exact values), whose palindromic characteristic polynomial $\lambda^4 - 6340\lambda^3 + 67974\lambda^2 - 6340\lambda + 1$ factors over $\mathbb{Q}(\sqrt{3})$ into two reciprocal quadratic sectors. The family splits into two period ladders: the weight-5 descent $(q_n, s_n, \hat{P}_n)$ rides the recurrence, the primitive (P_n, I'_n) does not. Using the filtration identity $I_n = I'_n + \zeta(2)\, I''_n$ and the new exact $s_3, \hat{P}_3$, we determine $P_3 = 23478462179525/(2^5 3^7)$ by a denominator-snap argument that requires *no* integral evaluation, verified independently (99.75% series climb; rigorous tail bound). Consequently $\mathrm{den}(P_3) \mid d_3^7$: the decisive 3-adic cell shows no denominator anomaly. Measuring decay rates against the irrationality threshold e^{-7} shows both projections ride the sector-B rate 0.0939: the symmetric family cannot prove "one of $\zeta(5), \zeta(7)$ is irrational"; the $\zeta(2)$-elimination costs a factor ≈ 594 in decay, the archimedean mirror of the Betti lattice cost. On the weight-5 arithmetic side, the primes $p \le n$ of the central-binomial divisibility $\binom{2n}{n} \mid 6A_n$ (Phase 2) are reduced to a single new congruence: a vector-valued *Dwork descent* $p^5(p_n/q_n) \equiv p_a/q_a \bmod p^{2-\kappa}$, $a = \lfloor n/p \rfloor$, whose constants are the $n = 1$ ladder ratios $29/28 = p_1/q_1$, $101/84 = \widetilde{p}_1/q_1$; the general inequality $\mathrm{ord}_p(p_n) \ge \kappa - 5L$ is verified with zero failures. Every claim below is labeled proven, verified, or conjectural.

1 Setting and summary of status

Brown–Zudilin [1] construct cellular integrals I_n on $\overline{\mathcal{M}}_{0,N}$; at $N = 10$, the totally symmetric convergent cell $\sigma = (10, 2, 4, 1, 6, 3, 8, 5, 9, 7)$ yields a rank-4 motive with weight-graded pieces $\mathbb{Q}(0), \mathbb{Q}(-2), \mathbb{Q}(-5), \mathbb{Q}(-7)$ and linear forms

$$I'_n = \tfrac{75}{4} q_n \zeta(7) - 3 s_n \zeta(5) - P_n, \qquad I''_n = -9 q_n \zeta(5) + 2 s_n \zeta(3) - \hat{P}_n, \qquad I_n = I'_n + \zeta(2)\, I''_n,$$

with printed anchors $q_n = 1, 61, 52921$; $s_n = 0, 300, 261153$; $P_n = 0, 220, 6021219/32$; $\hat{P}_n = 0, 152, 535857/4$ for $n = 0, 1, 2$ [1]. The leading coefficients q_n are computable for all n by the McCarthy–Osburn–Straub diagonal construction [3], verified against the anchors and extended here to $n \le 104$.

Status ledger. Throughout, *proven* = complete proof; *verified* = exact finite computation, no proof for all n; *conditional* = exact consequence of named hypotheses.

Result	Status	Where
(W): $w_n \equiv \widetilde{w}_n \equiv 0\ (p),\ n < p \le 2n$	**proven** (all n)	[4]
Phase 1 of $\binom{2n}{n} \mid 6A_n$	**proven** (all n)	[4]
q_n recurrence (order 4, deg 19)	verified (74 terms)	§2
Descent membership of $s_n, \hat{P}_n$	conditional/verified	§3
$s_3, \hat{P}_3$ exact values	conditional on membership	§3
P_3 exact value	conditional (2 hypotheses)	§4
Sector measurement (decay rates)	verified/measured	§5
Phase-2 reduction to (DWORK)	verified (0 failures)	§6
(DWORK) itself	**open** (new)	§6

2 The exact recurrence and its two sectors

Finding 1 (verified)**.** The sequence q_n of the totally symmetric $M_{0,10}$ cell satisfies a linear recurrence of order 4 with polynomial coefficients of degree 19 (leading/trailing coefficients palindromic, $7381728 \cdot [1, -6340, 67974, -6340, 1]$), reconstructed from 105 terms by 3-prime CRT ($\approx 2^{125}$) with exact rational lifting, and certified to annihilate all 74 exact values of q_n (all 70 testable offsets vanish identically; independently re-verified). The characteristic polynomial

$$\chi(\lambda) = \lambda^4 - 6340\lambda^3 + 67974\lambda^2 - 6340\lambda + 1 = \big(\lambda^2 - a\lambda + 1\big)\big(\lambda^2 - b\lambda + 1\big), \qquad a, b = 3170 \pm 1824\sqrt{3},$$

factors over $\mathbb{Q}(\sqrt{3})$ (check: $a + b = 6340$, $2 + ab = 67974$). The four roots form two reciprocal pairs (Poincaré-duality pairing): *sector A* $\{6329.2605\ldots,\ 1.57996\ldots \times 10^{-4}\}$ and *sector B* $\{10.6454\ldots,\ 0.0939374\ldots\}$.

The order matches the motivic rank ($\zeta(3)$: order 2; $\zeta(5)$: order 3; $\zeta(7)$: order 4), while the degree jump ($9 \to 19$ from weight 5) shows arithmetic complexity growing faster than rank. Annihilation of finitely many terms is not a proof that the operator governs q_n for all n; a creative-telescoping certificate was attempted on two representations (an 8-variable low-coupling window model and the 4-fold hypergeometric-sum form of §4) and in both cases hit the geometric elimination wall (successive eliminations $\ge$ 3h each); checkpoints are preserved. The recurrence's status therefore remains *verified*, with overwhelming structural support (motivic order match, palindromy, two independent window representations agreeing on all computed terms).

3 Two period ladders

Finding 2 (verified + conditional)**.** The weight-5 descent data $(q_n, s_n, \hat{P}_n)$ — equivalently the form I''_n — ride the order-4 operator: the three known values of each, augmented by the operator's trailing-coefficient degeneracy at the initial offset (which also re-derives $q_3 = 94357501$ from q_0, q_1, q_2, an internal consistency check), determine

$$s_3 = \frac{1396906795}{3}, \qquad \hat{P}_3 = \frac{232175579999}{972} \quad (972 = 2^2 3^5).$$

The primitive data (P_n, I'_n) do *not* satisfy the operator: propagation is inconsistent (the naive P_3 candidate even acquires a spurious prime factor 107). Weight seven thus splits into two holonomic species — the descended and the primitive ladder — which weight five does not separate.

The values $s_3, \hat{P}_3$ are exact *conditional on descent membership* (that the actual $s_n, \hat{P}_n$ satisfy the operator for all n); membership is verified at all available orders and is implied by the motivic rank structure, but a period-level telescoping certificate is the outstanding closure (see §7).

4 The value of P_3 by denominator snap

The full cellular integral admits the exact all-positive 4-fold series representation of [6] (verified against the anchors), hence $I_n > 0$ and $I_{n+1} < I_n$ rigorously. The observed values collapse: $I_0 = 3.5554\ldots$, $I_1 = 3.2071\ldots \times 10^{-5}$, $I_2 = 1.05313\ldots \times 10^{-9}$ — the full form is a near-perfect cancellation $I' \approx -\zeta(2)I''$ (four orders at $n = 2$).

Proposition 3 (snap; conditional on (i)–(ii))**.** *Assume* (i) $\mathrm{den}(P_3) \mid 12\, d_3^7$ *and* (ii) *descent membership (so $s_3, \hat{P}_3$ are as above). Then*

$$P_3 = \frac{23478462179525}{69984}, \qquad \mathrm{den}(P_3) = 69984 = 2^5\, 3^7.$$

Proof (no integral evaluated). By the filtration identity, $P_3 = \left[\frac{75}{4}q_3\zeta(7) - 3s_3\zeta(5) + \zeta(2)I_3''\right] - I_3$, where the bracket is computable to arbitrary precision from (ii) and $I_3'' = -9q_3\zeta(5) + 2s_3\zeta(3) - \hat{P}_3$. The grid $(1/12d_3^7)\mathbb{Z}$ of hypothesis (i) has spacing $h = 2.977 \times 10^{-7}$, and rigorously $0 < I_3 < I_2 = 1.053 \times 10^{-9} < h/2$; hence P_3 is the unique grid point within $h/2$ of the bracket, which is the stated rational. No numerical precision enters: positivity and monotonicity alone pin the grid point. $\square$

Independent verification. The implied value $I_3 = 5.6299224184893 \times 10^{-14}$ was tested against the all-positive series: partial sums (rigorous lower bounds) climb to 99.75% of it without ever exceeding it, and power-law extrapolation lands within 0.04%; a proved separable majorant gives the rigorous bound $I_3 \leq 3.09 \times 10^{-9} < h/2$, making the grid choice unconditional given (i). Consistency locks: the residual is positive as required; $I_3/I_2 = 5.35 \times 10^{-5}$ continues the ratio sequence 9.0×10^{-6}, 3.28×10^{-5} monotonically toward the sector-A root 1.58×10^{-4}, as duality predicts for the full form; a wrong grid hypothesis would land within 5.6×10^{-14} of a grid point with probability $\sim 2 \times 10^{-7}$.

Corollary 4 (conditional)**.** $\mathrm{den}(P_3) = 2^5 3^7$ *divides* $d_3^7 = 2^7 3^7$*: the naive inclusion* $d_3^7 P_3 \in \mathbb{Z}$ *holds at* $n = 3$*, with* ord_3 *tight and* ord_2 *slack* 2*. At its first* 3*-adic opportunity the weight-7 primitive ladder shows* no 12*-type correction — consistent with the* $n \leq 2$ *data and with* $\mathrm{den}(\hat{P}_3) = 2^2 3^5$ *on the descent ladder: the anomalous correction factor measured at weight 5 [5] does not recur in this family through* $n = 3$*.*

5 Sector measurement and the Diophantine verdict

For a weight-7 form with denominator growth $d_n^{7+o(1)}$, irrationality-type conclusions require decay below $e^{-7} = 9.119 \times 10^{-4}$. The two sub-unit roots straddle this threshold: sector A's 1.58×10^{-4} passes (margin $e^{1.753}$), sector B's 0.0939 fails by two orders.

Finding 5 (measured)**.** Propagating I_n'' through the exact operator at 300-digit precision from its four exact initial values gives ratios increasing monotonically $0.0042,\ 0.0181,\ 0.0306,\ \ldots,\ 0.0833$ ($n = 29$) toward 0.0939: the *descended* ladder rides sector B. The filtration then locks the primitive projection: since I_n (sector A, ratios $9.0 \times 10^{-6}, 3.3 \times 10^{-5}, 5.4 \times 10^{-5} \to 1.58 \times 10^{-4}$) is negligible against $\zeta(2)I_n''$, we get $I_n' \approx -\zeta(2)I_n''$, and indeed $I_3'/I_2' = 0.0306408$ against $I_3''/I_2'' = 0.0306408$ (difference 2.8×10^{-8}). *Both projections ride sector B.*

Corollary 6 (conditional on descent membership)**.** *The totally symmetric* $\zeta(7)$ *family cannot prove "at least one of* $\zeta(5), \zeta(7)$ *is irrational" under a* d_n^7*-type denominator cost: the required decay* e^{-7} *is missed by the factor* $0.0939/e^{-7} \approx 103$*. The* $\zeta(2)$*-elimination costs a decay factor*

$0.0939/1.58\times10^{-4}\approx 594$ — *an archimedean analogue of the Betti elimination cost of [5]. The surviving Diophantine route is asymmetric: optimize $\mathcal{M}(a)=-\log\rho_{\mathrm{proj}}(a)-\kappa(a)$ over the parameter orbit (decay and denominator cost move together; positive $\mathcal{M}$ is the Diophantine region).*

6 Phase 2 of the central-binomial law: reduction to a Dwork congruence

We now return to weight 5. With Zudilin's sequences $q_n, p_n, \widetilde{p}_n$ [2] and $A_n = 2d_n^5 p_n \in \mathbb{Z}$, the corrected symmetric law $12d_n^5 P_n \in \mathbb{Z}$ is equivalent to $\binom{2n}{n} \mid 6A_n$, i.e. $\mathrm{ord}_p(p_n) \geq \kappa - 5L$ at every prime $p \geq 5$, where $\kappa = \mathrm{ord}_p\binom{2n}{n}$ (Kummer's carry count) and $L = \mathrm{ord}_p\, d_n = \lfloor \log_p n\rfloor$. Phase 1 ($n < p \leq 2n$, $L = 0$) is proven in [4]. For $p \leq n$ the following structure was discovered and verified in this campaign.

Finding 7 (verified; 272 descents, 46 band pairs, 0 failures)**.** Let $a = \lfloor n/p \rfloor$. Then

$$p^5\,\frac{p_n}{q_n} \;\equiv\; \frac{p_a}{q_a}, \qquad p^3\,\frac{\widetilde{p}_n}{q_n} \;\equiv\; \frac{\widetilde{p}_a}{q_a} \pmod{p^k}, \qquad k \geq 2-\kappa \text{ (typical } 3-\kappa),$$

for all tested (n,p), $p \geq 5$, $n \leq 45$. The digit constants are exact ladder values — in particular

$$\frac{p_1}{q_1} = \frac{29}{28}, \qquad \frac{\widetilde{p}_1}{q_1} = \frac{101}{84}, \qquad \frac{p_2}{q_2} = \frac{24289}{23424},$$

identified first as the unique low-height fit to the measured p-adic digits and then confirmed exactly. The first correction digit is antisymmetric in $r = n - ap$ about $(p-1)/2$ — the derivative signature of a Dwork $F(x)/F(x^p)$ congruence. Plain Lucas congruences $q_{ap+b} \equiv q_a q_b\ (p)$ *fail* at every tested prime: the operative structure is a Frobenius descent of the ratio of solutions, not a digit product. At exceptional primes dividing a digit-constant denominator (e.g. $7 \mid 28$), the ratio statement degrades but the combined integer invariant $\mathrm{ord}_p(p_n) \geq \kappa - 5L$ is preserved in every tested case.

Proposition 8 (assembly; proven modulo (DWORK))**.** *Assume, for all $p \geq 5$ and n,* (DWORK)*: the descent congruence of Finding 7 with $k \geq 1$ (in the denominator-free vector form: cross-multiplied by q_a, for the pair $(p_n, \widetilde{p}_n)$). Then, using* $\mathrm{ord}_p(q_n) \geq \kappa$ *(free: $Q_n = \pm q_n/\binom{2n}{n}$ is Brown–Zudilin's manifestly integral double sum) and iterating the descent through the L base-p digit levels (each level costing the d_n^5-reserve exponent* 5*), one obtains* $\mathrm{ord}_p(p_n) \geq \kappa - 5L$ *for all $p \geq 5$: Phase 2 of the central-binomial law. Combined with Phase 1 [4] this yields $\binom{2n}{n} \mid 6A_n$ and hence $12\,d_n^5 P_n \in \mathbb{Z}$ for all n.*

The inequality $\mathrm{ord}_p(p_n) \geq \kappa - 5L$ itself is verified with zero failures for all $p \geq 5$, $n \leq 45$ — including the 206 zero-slack tight cases of the earlier ledger, which the descent now explains: the d_n^5 reserve pays exactly 5 orders per digit level, and κ carries the Kummer carries.

Literature status and proof program. A targeted audit (with searches of the primary sources) indicates (DWORK) is *not* covered off-the-shelf: Delaygue–Rivoal–Roques [7] treat hypergeometric coefficient sequences and mirror-map quotients, not harmonic-companion ratios; Malik–Straub [8] prove scalar Lucas-type results and leave stronger Dwork congruences open; Gorodetsky's constant-term methods strengthen the scalar toolkit for q_n only. The planned proof route: (1) restate (DWORK) denominator-free in the period vector $(q_a, p_a, \widetilde{p}_a)$, so exceptional primes become lattice bookkeeping; (2) prove the $a = 1$ band by splitting the very-well-poised summand into p-blocks and expanding factorial and harmonic blocks through weight 5 (Wolstenholme-block cancellations);

(3) treat $(p_n, \widetilde{p}_n)$ simultaneously, exposing a triangular Frobenius matrix; (4) generalize in a. The $a = 0$ instance of (DWORK) is exactly the proven Phase-1 congruence — a consistency anchor. A period-level telescoping certificate for the weight-7 family (§2) would additionally supply the period-matrix realization placing (DWORK) inside matrix Dwork theory.

7 Outlook

Three closures would upgrade the night's conditional results to theorems: (1) a creative-telescoping certificate (or alternative proof) that the order-4 operator governs the weight-7 family at period level — closing descent membership, hence $s_3, \hat{P}_3$, hence hypothesis (ii) of the snap; (2) a proof of the denominator-grid hypothesis (i), e.g. by the Krattenthaler–Rivoal-type arithmetic for the weight-7 forms; (3) a proof of (DWORK), completing the corrected Brown–Zudilin law $12d_n^5 P_n \in \mathbb{Z}$. Independent of all three, the sharpest new open object is the vector-valued Dwork descent itself: one congruence family, verified across 272 instances, whose archimedean limit is the classical convergence $p_n/q_n \to \zeta(5)$ and whose p-adic content is a self-similarity under base-p digit shift. One ratio, all completions, one mechanism.

Provenance

Results obtained 17–18 July 2026 in a collaborative session: coordination, reductions and proofs as cited; exact computation by fleet agents; independent review by a second model (in particular the conditional-rigor form of Proposition 3 and the literature audit of §6). Complete scripts and exact data in the repository (`worthiness/`): recurrence and terms (`zeta7_q_recurrence.json`, `zeta7_lc_terms.txt`), snap and verification (`ZETA7_P3_SNAP.md`, `ZETA7_P3_VERIFICATION.md`, `zeta7_p3_*.py`), sector measurement (`zeta7_sector_measurement.py`), Phase-2 program (`PHASE2_BAND_BLUEPRINT.md`, `PHASE2_BAND_V.md`, `lemma_cb_band_v*.py`).

References

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[4] *A Frobenius certificate for central-binomial cancellation in Zudilin's symmetric* $\zeta(5)$ *construction*, companion note, 2026 (`worthiness/cb_certificate.tex`).

[5] *Sharp denominator corrections for Brown–Zudilin cellular approximations*, companion paper, 2026 (`paper/denominators.tex`).

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[8] A. Malik, A. Straub, *Divisibility properties of sporadic Apéry-like numbers*, Res. Number Theory (2016).